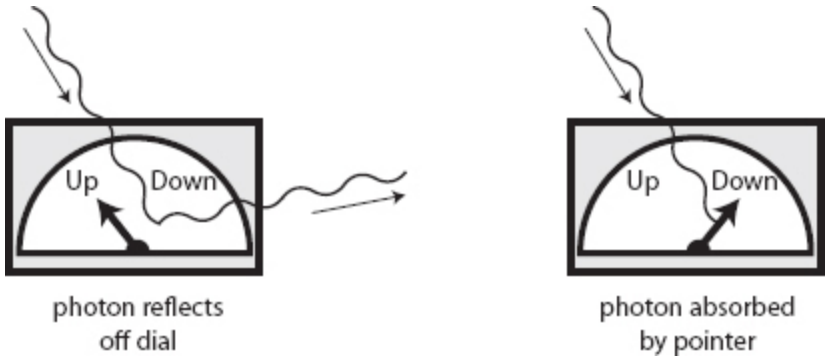
crucial to making sense of quantum mechanics. It explains once and for all why wave functions seem to collapse when you measure quantum systems—and indeed what a "measurement" really is.

We know there is only one wave function, the wave function of the universe. But when we're talking about individual microscopic particles, they can settle into quantum states where they are unentangled from the rest of the world. In that case, we can sensibly talk about "the wave function of this particular electron" and so forth, keeping in mind that it's really just a useful shortcut we can employ when systems are unentangled with anything else.

With macroscopic objects, things aren't that simple. Consider our spin-measuring apparatus, and let's imagine we put it in a superposition of having measured spin-up and spin-down. The dial of the apparatus includes a pointer that is pointing either to Up or to Down. An apparatus like that doesn't stay separate from the rest of the world. Even if it looks like it's just sitting there, in reality the air molecules in the room are constantly bumping into it, photons of light are bouncing off of it, and so on. Call all that other stuff—the entire rest of the universe—the *environment*. In ordinary situations, there's no way to stop a macroscopic object from interacting with its environment, even if very gently. Such interactions will cause the apparatus to become entangled with the environment, for example, because a photon would reflect off the dial if the pointer is in one position, but be absorbed by it if the pointer is pointing somewhere else.



So the wave function we wrote down above, where an apparatus became entangled with a qubit, wasn't quite the whole story. Putting the environment states in curly braces, we should have written